Exact Algebra of Retained Continuation

Native Binary Rings, Four-Involution Field Dynamics, and the Monodromy of Multi-Reader Closures

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November 2026

The physical and computational universe documented across the NEXUS research corpus is the output of deterministic binary transitions executing on a fixed, self-dual memory map: the native binary ring R_n. Across the individual investigations—spanning the S_3 compiler triad Γ, the three-point route switch Q_{op}, the Boolean Möbius difference calculus, single-reader writable-field closures, and multi-reader composite interactions—a single overarching mathematical structure governs every regime. That structure is the algebraic interplay between bitwise XOR translation on $\mathbb{F}_2^n$ and the piecewise-linear native partner involution P_n, which acts as bit-reversal J_n on non-palindromic words and bitwise complement $C_n(x) = x \oplus u$ (where $u = 2^n - 1$) on palindromes. When the ring R_n is queried as a static mold without field modification, the triad Γ and route switch Q_{op} execute reversible continuation with zero stack growth, while the Boolean Möbius transform M conjugates circular address shift T directly into the local nilpotent difference stack $MD^{(r)} = Sa^{(r)}$ across every bit plane.

Allowing $m \geq 1$ readers to write their carried partner transformations back into the shared offset field $D = V \oplus M$ elevates the system from reversible continuation to retained continuation $F^{(m)}$. In this paper, we establish the unified operator algebra of retained continuation and prove four new structural theorems verified by independent executable code. First, we prove the **Four-Involution Factorization Theorem**: for any number of simultaneous readers $m \geq 1$ and arbitrary multi-reader address collisions, the coupled forward step factors into four atomic state-space involutions, $F^{(m)} = \mathcal{M} \circ \mathcal{W} \circ \Pi \circ \mathcal{R}$, yielding an unconditional, order-independent inverse $(F^{(m)})^{-1} = \mathcal{R} \circ \Pi \circ \mathcal{W} \circ \mathcal{M}$ and conserving the 1-bit shell-lamp charges $\mathcal{L}_k$ across all shells. Second, we show that the transition from single-shell confinement to cross-shell interaction is governed by the polynomial degree of P_n over $\mathbb{F}_2$: on any 8-word single-shell carrier $K(w) \cong \mathbb{F}_2^3$, $P_8(x) = A(x) \oplus u$ is an exact $\mathbb{F}_2$-affine involution, conserving a 3-bit vector charge $\mathcal{Q} = B_D \oplus A(p_A \oplus p_B)$ that locks the readers' relative XOR separation $p_A(t) \oplus p_B(t) = p_A(0) \oplus p_B(0)$ at every vacuum return $D_t = 0$.

Third, by combining the 3-bit affine charge $\mathcal{Q}$ with the 64-element equivariance group $\mathcal{G}_{64}$ of the two-reader machine, we construct the **Vacuum Monodromy Compiler** on the 64 empty-field seeds of the 8-word center carrier. Because every vacuum return of a seed $S_0(\sigma)$ lands on a $\mathcal{G}_{64}$-symmetry image of another seed

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$S_0(\pi(\sigma))$, tracing the monodromy cocycle across the first-vacuum return table resolves the exact periods, total vacuum visits, and half-period involutions for all 64 seeds—closing all 28 trajectories that remained open at the 300,000,000-tick census horizon into two exact macro-orbits of periods 310,826,368 (16 vacuum returns) and 515,455,952 (28 vacuum returns). Fourth, on the 32-word two-shell carrier $W \cong \mathbb{F}_2^5$, we prove that the native partner becomes a degree-2 Boolean polynomial $P_5(x) = J_5(x) \oplus (1 \oplus \ell_0(x))(1 \oplus \ell_1(x))u$, whose quadratic cross-term $\ell_0(x)\ell_1(x)u$ breaks the 3-bit charge Q. However, for the shared-start cross-shell configuration $A = (0, e_R), B = (0, o_R)$, shell-exchange symmetry Σ_{shell} annihilates the quadratic u-kick under the diagonal map $\delta = I \oplus S_{\text{shell}}$, restoring an exact cross-shell conservation law $Q_{\text{shell}} = B_D(t) \oplus J_5(p_A(t) \oplus p_B(t)) = 0$ and proving by exhaustive induction over all 2,528 reachable transitions that the compound is permanently confined to 28 words, 288 reader-carry states, and 44 independent field bits.

1. The Native Binary Ring, Facing Geometry, and Symmetric Group Generation

1.1 Construction of the Native Ring and the Universal Facing Law

Let $\mathcal{W}_n = \mathbb{F}_2^n \cong \{0,1,\dots,2^n-1\}$ denote the n-bit word space equipped with bitwise addition (XOR, $\oplus$), all-ones word $u = 2^n - 1$, and bitwise complement $C_n(x) = x \oplus u$. For any word $x = \sum_{k=0}^{n-1} x_k\, 2^k \in \mathcal{W}_n$, bit-reversal is the linear involution $J_n(x) = \sum_{k=0}^{n-1} x_k\, 2^{n-1-k}$, and the palindrome subspace is the fixed-point subspace $\mathrm{Pal}_n = \{x \in \mathcal{W}_n : J_n(x) = x\}$, which has dimension $\lfloor (n+1)/2 \rfloor$ and cardinality $|\mathrm{Pal}_n| = 2^{\lfloor (n+1)/2 \rfloor}$.

The native partner $P_n : \mathcal{W}_n \to \mathcal{W}_n$ is defined piecewise by:

$$P_n(x) = \begin{cases} J_n(x), & \text{if } J_n(x) \neq x, \\ C_n(x) = x \oplus u, & \text{if } J_n(x) = x. \end{cases}$$

Because $J_n(u) = u$ and J_n is linear, $J_n(x \oplus u) = J_n(x) \oplus u$, so P_n maps Pal_n onto Pal_n without fixed points and $\mathcal{W}_n \backslash \mathrm{Pal}_n$ onto itself without fixed points. Thus $P_n^2 = I$ and $P_n(x) \neq x$ for all $x \in \mathcal{W}_n$, partitioning $\mathcal{W}_n$ into $h = 2^{n-1}$ disjoint partner pairs $\{x, P_n(x)\}$.

To construct the native binary ring R_n of length $N = 2^n$, let $r_0 < r_1 < \cdots < r_{h-1}$ be the $h = 2^{n-1}$ pair representatives satisfying $r_k < P_n(r_k)$ in standard integer order. The ring lists the first $h-1$ representatives in ascending order, swaps the last representative $m = r_{h-1}$ with its partner $P_n(m)$ at the central crease (the seam between addresses $h-1$ and h), and lists the remaining partners in descending representative order: $R_n = [\, r_0,\, r_1,\, \dots,\, r_{h-2},\, P_n(r_{h-1}),\, r_{h-1},\, P_n(r_{h-2}),\, \dots,\, P_n(r_1),\, P_n(r_0)\,]$.Because P_n is an involution, this construction satisfies the universal facing law at every address $a \in \{0,1,\dots,N-1\}$:

$$R_n[N-1-a] = P_n(R_n[a]).$$

We denote the inverse address lookup map by $\mathrm{READ}_n : \mathcal{W}_n \to \{0,1,\dots,N-1\}$, so that $\mathrm{READ}_n(R_n[a]) = a$ and $R_n[\mathrm{READ}_n(w)] = w$.

1.2 The Seam Theorem, the Center Carrier, and the Innermost Diamond

At even width $n = 2r$ ($r \geq 1$), let $e_R = 2^{r-1}$ and $e_L = 2^r$ denote the two center-bit generators, and let $s = e_L \oplus e_R = 3 \cdot 2^{r-1}$ denote the center-pair word.

Theorem 1.1 (Even-Width Seam and Center Carrier Address Law). *For every even width $n = 2r \geq 2$ with $h = 2^{n-1}$ and $u = 2^n - 1$, the last representative is $r_{h-1} = u \oplus e_L$, and the seam words at addresses $h-1$ and h are:*

$$R_n[h-1] = u \oplus e_R, \qquad R_n[h] = u \oplus e_L, \qquad R_n[h-1] \oplus R_n[h] = s = e_L \oplus e_R.$$

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Furthermore, for every even width $n = 2r \geq 4$, the 8-word center carrier $L = \mathrm{span}_{\mathbb{F}_2}\{u, e_R, e_L\} = \{0, e_R, e_L, s, u \oplus s, u \oplus e_R, u \oplus e_L, u\}$ occupies the exact closed-form ring addresses:

$$\mathrm{READ}_n(0) = 0, \quad \mathrm{READ}_n(e_R) = 2^{r-1}, \quad \mathrm{READ}_n(s) = 3 \cdot 2^{r-1} - 1, \quad \mathrm{READ}_n(u \oplus e_R) = 2^{2r-1} - 1,$$
$$\mathrm{READ}_n(u \oplus e_L) = 2^{2r-1}, \quad \mathrm{READ}_n(u \oplus s) = 2^{2r} - 3 \cdot 2^{r-1}, \quad \mathrm{READ}_n(e_L) = 2^{2r} - 1 - 2^{r-1}, \quad \mathrm{READ}_n(u) = 2^{2r} - 1.$$

Proof. Any word $x > u \oplus e_L = (2^{r-1} - 1)2^{r+1} + (2^r - 1)$ has its upper r bits all equal to 1, so $x = (2^r - 1)2^r + y$ with $0 \leq y \leq 2^r - 1$. If $y = 2^r - 1$, then $x = u \in \mathrm{Pal}_n$, whose partner is $P_n(u) = 0 < u$. If $y < 2^r - 1$, then $J_r(y) < 2^r - 1$, so $J_n(x) = J_r(y)2^r + (2^r - 1) < x$, meaning $P_n(x) = J_n(x) < x$. Thus no word strictly greater than $u \oplus e_L$ satisfies $x < P_n(x)$. Conversely, $J_n(u \oplus e_L) = u \oplus e_R = u - 2^{r-1} > u - 2^r = u \oplus e_L$, so $r_{h-1} = u \oplus e_L$ is the largest representative. By the seam swap in R_n, $R_n[h-1] = P_n(r_{h-1}) = u \oplus e_R$ and $R_n[h] = r_{h-1} = u \oplus e_L$.

For the remaining addresses in L at $n = 2r \geq 4$: every nonzero word $x < 2^{r-1} = e_R$ has its upper $r + 1$ bits zero, so $J_n(x)$ has a 1 in its upper $r - 1$ bits and thus $J_n(x) \geq 2^{r+1} > x$. Hence all 2^{r-1} words in $\{0,1, \ldots, 2^{r-1} - 1\}$ are representatives, placing $e_R = 2^{r-1}$ (which is also a representative since $J_n(e_R) = e_L = 2^r > e_R$) at $\mathrm{READ}_n(e_R) = 2^{r-1}$. For $s = 3 \cdot 2^{r-1}$, every word $x \in \{1, \ldots, 3 \cdot 2^{r-1} - 1\}$ has upper r-bit prefix $x_{\mathrm{hi}} \in \{0,1\}$. If $x_{\mathrm{hi}} = 0$ and $x > 0$, then $J_n(x) \geq 2^r > x$. If $x_{\mathrm{hi}} = 1$, then $x = e_L \oplus y$ with $0 \leq y < 2^{r-1}$; for $y \neq 0$, $J_n(x) \geq 2^{r+1} > x$, and for $y = 0$, $x = e_L = 2^r$, where $P_n(e_L) = e_R < e_L$. Thus e_L is the single non-representative below s, giving $\mathrm{READ}_n(s) = 3 \cdot 2^{r-1} - 1$. The partner addresses follow immediately from the facing law $\mathrm{READ}_n(P_n(w)) = 2^{2r} - 1 - \mathrm{READ}_n(w)$. ■

At every even width $n = 2r \geq 6$, let $d = r - 2$ and define the anchor word $m^* = 1^d00011^d = ((2^d - 1) \ll (d + 4)) \mid (1 \ll d) \mid (2^d - 1)$. The four words of the coset $m^* \oplus \{0, e_R, s, e_L\} = \{1^d00011^d, 1^d00111^d, 1^d01111^d, 1^d01011^d\}$ sit at the exact width-independent seam offsets:

$$[\mathrm{READ}_n(m^*), \mathrm{READ}_n(m^* \oplus e_R), \mathrm{READ}_n(m^* \oplus s), \mathrm{READ}_n(m^* \oplus e_L)] - h = [-6, -4, -2, -3].$$

For example, on the 64-address ring R_6 ($h = 32$), the eight words of the center carrier L span the origin at address 0, the terminus at address 63, the central seam between addresses 31 and 32, and two symmetric facing chords connecting addresses 4 and 59 (e_R and e_L) and addresses 11 and 52 (s and $u \oplus s$), while the innermost diamond occupies the fixed addresses 26, 28, 30, and 29. At odd widths $n = 2r + 1$, the single center bit $c = 2^r \in \mathrm{Pal}_n$ sits in the opening representative run at $\mathrm{READ}_n(c) = c$, while the seam words are $u \oplus 2^{r-1}$ and $u \oplus 2^{r+1}$, differing by $2^{r+1} \oplus 2^{r-1}$. Table 1 summarizes the verified seam geometry and center carrier addresses across widths 2 through 14.

Width n	Ring Size N=2n	Seam Pair Rn [h−1]|Rn[h]	Seam XOR Difference	Center Carrier L Addresses (0,eR,s,u⊕eR,u⊕eL,u⊕s,eL,u) or Odd Center (c,u⊕c)	Innermost Diamond Offsets from h
2	4	10 | 01	11	(0, 2, 3, 1, 2, 0, 1, 3)	—

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Width n	Ring Size N=2n	Seam Pair Rn [h−1]|Rn[h]	Seam XOR Difference	Center Carrier L Addresses (o,eR,s,u⊕eR,u⊕eL,u⊕s,eL,u) or Odd Center (c,u⊕c)	Innermost Diamond Offsets from h
3	8	110 | 011	101	c=010 at 2, u^c=101 at 5	—
4	16	1101 | 1011	0110	(0, 2, 5, 7, 8, 10, 13, 15)	—
5	32	11101 | 10111	01010	c=00100 at 4, u^c=11011 at 27	—
6	64	111011 | 110111	001100	(0, 4, 11, 31, 32, 52, 59, 63)	[-6, -4, -2, -3]
7	128	1111011 | 1101111	0010100	c at 8, u^c at 119	—
8	256	11110111 | 11101111	00011000	(0, 8, 23, 127, 128, 232, 247, 255)	[-6, -4, -2, -3]
10	1,024	1111101111 | 1111011111	0000110000	(0, 16, 47, 511, 512, 976, 1007, 1023)	[-6, -4, -2, -3]
12	4,096	111111011111 | 111110111111	000001100000	(0, 32, 95, 2047, 2048, 4000, 4063, 4095)	[-6, -4, -2, -3]
14	16,384	11111110111111 | 11111101111111	00000011000000	(0, 64, 191, 8191, 8192, 16192, 16319, 16383)	[-6, -4, -2, -3]

1.3 Discrepancy Operator and Generation of the Full Symmetric Group

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Let $T: \mathcal{W}_n \to \mathcal{W}_n$ be the native ring successor defined by $T(R_n[a]) = R_n[(a+1) \bmod 2^n]$. Because $R_n[N-1-a] = P_n(R_n[a])$, conjugating T by P_n reverses the ring traversal:

$$P_n T P_n = T^{-1}, \qquad \begin{aligned} P_n(R_n[a]) &= R_n[(-a-1) \bmod 2^n], \\ &= R_n[(-a) \bmod 2^n], \end{aligned} \qquad \begin{aligned} (TP_n)(R_n[a]) \\ (P_n T)(R_n[a]) = R_n[(-a-2) \bmod 2^n]. \end{aligned}$$

Consequently, $\langle T, P_n \rangle$ is the dihedral group of order 2^{n+1} (the symmetry group of the 2^n-gon) acting on the 2^n-cycle R_n, and the two reflections TP_n and $P_n T$ are conjugate via $TP_n = T(P_n T)T^{-1}$.

When bitwise complement C_n is adjoined to $\langle T, P_n \rangle$, the generated permutation group $\langle T, P_n, C_n \rangle$ depends critically on whether P_n and C_n coincide:

1. **Width $n = 2$ (Dihedral Collapse):** On $R_2 = [0,2,1,3]$, the non-palindromes are $1 = (01)_2$ and $2 = (10)_2$, for which $J_2(1) = 2 = C_2(1)$. Thus $P_2 = C_2$ identically on all 4 words, so $\langle T, P_2, C_2 \rangle = \langle T, P_2 \rangle \cong D_4$ has order 8 (strictly smaller than $|S_4| = 24$).
2. **Widths $n \geq 3$ (Full Symmetric Group Generation):** For all $n \geq 3$, the composite operator $P_n C_n$ satisfies:

$$(P_n C_n)(x) = \begin{cases} x, & \text{if } x \in \mathrm{Pal}_n, \\ J_n(x) \oplus u, & \text{if } x \notin \mathrm{Pal}_n. \end{cases}$$

Thus $P_n C_n$ is a non-identity involution fixing exactly the words on which $P_n = C_n$: the $2^{\lfloor (n+1)/2 \rfloor} \geq 4$ palindromes and, at even n, the $2^{n/2}$ anti-palindromes (for which $J_n(x) \oplus u = x$). It swaps the remaining $2^n - 2^{\lfloor (n+2)/2 \rfloor}$ words in pairs $\{x, J_n(x) \oplus u\}$, and these are exactly the R_3 carries of Theorem 3.1. In address coordinates on R_n, C_n shatters all dihedral block systems of $\langle T, P_n \rangle$, so $\langle T, P_n, C_n \rangle$ is primitive and contains the regular cyclic subgroup $\langle T \rangle$. By the classification of primitive groups with a regular cyclic subgroup (G. A. Jones, J. Group Theory 5 (2002) 403–407), a primitive group of degree $2^n \geq 8$ containing a 2^n-cycle is S_{2^n}, A_{2^n}, or, when $q = 2^n - 1$ is prime, $\mathrm{PGL}_2(q)$ acting on the projective line. The involution $P_n C_n$ fixes at least $2^{\lfloor (n+1)/2 \rfloor} \geq 4$ points, which excludes $\mathrm{PGL}_2(q)$, whose involutions fix at most 2 points; the successor T is a single 2^n-cycle and therefore an odd permutation, which excludes A_{2^n} ($P_n C_n$ itself is an even permutation). This yields $\langle T, P_n, C_n \rangle = S_{2^n}$. At $n = 3$, exhaustive breadth-first enumeration confirms $|\langle T, P_3, C_3 \rangle| = 40{,}320 = 8!$, and computational verification extends this identity through $n = 12$ (4,096 words). Every reversible permutation of an n-bit state space ($n \geq 3$) is therefore an executable word in $\{T, P_n, C_n\}$.

2. Reversible Continuation: The Compiler Triad, Route Switching, and Boolean Möbius Conjugacy

2.1 The One-Byte Compiler Circle on R_2

Before introducing a writable field, the native ring R_n operates as a zero-retention reversible compiler. At $n = 2$, reading the four 2-bit words of $R_2 = [00,10,01,11]$ sequentially produces the 8-bit stored circle $s = (00100111)_2 = 0x27$. Let T_8 act on $\mathbb{F}_2^8$ by cyclic left bit-shift $(T_8 s)_k = s_{(k+1) \bmod 8}$, let J_8 reverse the 8 bits, and let C_8 complement them. Direct evaluation establishes five foundational byte identities:

1. **Full-Adder Sum and Carry Extraction:** The two-point cyclic difference $(I \oplus T_8)s = (01101001)_2 = 0x69$ is the exact 8-row truth table of the 3-input full-adder sum $a \oplus b \oplus c$ for $(a, b, c) \in \{0,1\}^3$, while the corresponding 3-input majority carry $\mathrm{maj}(a, b, c) = (ab) \oplus (bc) \oplus (ca)$ is $(00010111)_2 = 0x17$, obtained from $s = 0x27$ by moving the single bit at index 2 to index 3.
2. **Reversal as Shift:** Reversing the 8-bit stream is an exact 5-step cyclic shift: $J_8(0x27) = T_8^5(0x27) = (11100100)_2 = 0xE4$.

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3. **Sheet Reading and the Innermost Diamond:** Applying complement-reversal yields the sheet byte $C_8J_8(\text{ox27}) = (00011011)_2 = \text{ox1B}$ (standard binary counting order 00,01,10,11), whose cyclic difference $(I \oplus T_8)(\text{ox1B}) = (00101101)_2 = \text{ox2D}$ encodes the four 2-bit states $00 \to 10 \to 11 \to 01$ of the innermost diamond.

4. **End-Around Carry Doubling Split:** Rotating the sheet byte $\text{ox1B} = 27$ through its 8 cyclic windows shifts the integer value by $v_{k+1} = 2v_k - 255 \cdot b_{\text{MSB}}$. Because the first four bits of ox1B (0001…) have $b_{\text{MSB}} = 0$ for three shifts, the first four windows double exactly ($27 \to 54 \to 108 \to 216$), whereas the next four windows (1011…) carry a leading 1 across the boundary and subtract the ones'-complement modulus 255 ($216 \to 177 \to 99 \to 198 \to 141 \to 27$).

5. **Anchor-Free Three-Point Inversion:** Whereas $I \oplus T_8$ has a 1-dimensional kernel ($\ker(I \oplus T_8) = \{\text{ox00}, \text{oxFF}\}$) and loses the global sheet polarity, the three-point route switch $Q_8 = I \oplus T_8 \oplus T_8^2$ satisfies $Q_8^8 = I + T_8^8 + T_8^{16} = I \pmod 2$, so $(I \oplus T_8 \oplus T_8^2)(\text{ox27}) = (11110101)_2 = \text{oxF5}$ reconstructs $\text{ox27} = Q_8^7(\text{oxF5})$ with zero anchor bits.

2.2 The S_3 Compiler Triad and Projective Line Partition

On any ring R_n, a read of value $v \in \mathcal{W}_n$ at address $a \in \{0, \dots, 2^n - 1\}$ binds the word $m = R_n[a]$ (WHERE), the value v (WHAT), and the difference $d = m \oplus v$ (ROUTE) via the compiler map:

$\Gamma(a, v) = (\text{READ}_n(v),\ v \oplus R_n[a])$.In the word-space triad coordinates (m, v, d) constrained to the $2n$-dimensional zero-sum manifold $\mathcal{M}_n = \{(m, v, d) \in \mathcal{W}_n^3 : m \oplus v \oplus d = 0\}$ of cardinality 4^n, Γ acts as the cyclic 3-shift:$\Gamma(m, v, d) = (v, d, m), \qquad \Gamma^3 = I, \qquad \Gamma^2 = \Gamma^{-1}$.

Because the route $d = m \oplus v$ is retained in the register triad, applying Γ twice unbinds the state with zero stack growth.

The action of Γ and the full symmetric group S_3 (permuting (m, v, d)) partitions the 4^n states of $\mathcal{M}_n$ into three exact geometric strata:

1. **The Origin (1 state):** $(0,0,0)$, fixed by all of S_3.

2. **Zero-Gap Boundary Triads ($3(2^n - 1)$ states):** The $2^n - 1$ cyclic Γ-orbits of the form $\{(x, x, 0), (x, 0, x), (0, x, x)\}$ for $x \in \mathcal{W}_n \backslash \{0\}$.

3. **Interior Projective Triads ($6L_n$ states):** Triads (m, v, d) where $m, v, d \in \mathcal{W}_n \backslash \{0\}$ are distinct. Each unordered 3-set $\{m, v, m \oplus v\}$ is a 2-dimensional $\mathbb{F}_2$-subspace minus zero—an exact projective line of $\text{PG}(n - 1, 2)$. Each line supports 6 ordered triads forming two oriented Γ-orbits of length 3 (oppositely oriented inner loops), giving:

$$4^n = 1 + 3(2^n - 1) + 3 \cdot \frac{(2^n - 1)(2^n - 2)}{3} = 1 + 3(2^n - 1) + 6L_n, \qquad L_n = \frac{(2^n - 1)(2^n - 2)}{6}.$$

At the Fano width $n = 3$ ($\text{PG}(2,2)$), there are $2^3 - 1 = 7$ nonzero points and $L_3 = 7$ projective lines. Through every nonzero WHERE point $m \in \{1, \dots, 7\}$ pass $2^{3-1} - 1 = 3$ projective lines $L_m^{(0)}, L_m^{(1)}, L_m^{(2)}$ that partition the remaining 6 nonzero words. While $\Gamma^3 = I$ cycles the three roles (m, v, d) inside each line $L_m^{(k)}$, the route-switch operator $Q_t(m, v, d) = (m, v \oplus t, d \oplus t)$ switches between lines holding m fixed. Choosing one point $v_0 \in L_m^{(0)} \backslash \{m\}$ and one point $v_1 \in L_m^{(1)} \backslash \{m\}$ defines the transverse Fano line $\{t_{01}, t_{12}, t_{20}\} = \{v_0 \oplus v_1, v_0, v_1\}$ satisfying $t_{01} \oplus t_{12} \oplus t_{20} = 0$, which cycles the three lines meeting at m

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in a closed 3-step outer loop $Q_{t_{20}}Q_{t_{12}}Q_{t_{01}} = I$, whereas applying a self-line switch $t = v_0 \in L_m^{(0)}$ collapses $(m, v_0, m \oplus v_0)$ directly into the zero-gap triad $(m, 0, m)$.

2.3 The Route-Switch Defect Law and Perimeter Line Coverage

Along the perimeter of the native ring R_n, lift each directed edge $x_a \to x_{a+1}$ (where $x_a = R_n[a]$ and indices are read modulo $N = 2^n$) to the triad $E(a) = (x_a, x_{a+1}, x_a \oplus x_{a+1})$. One step of the compiler Γ predicts the state $\Gamma E(a) = (x_{a+1}, x_a \oplus x_{a+1}, x_a)$, whereas the next perimeter edge of the ring is $E(a+1) = (x_{a+1}, x_{a+2}, x_{a+1} \oplus x_{a+2})$. Subtracting the compiler prediction from the actual ring edge yields the route-switch transition:

$$E(a+1) = Q_{t_a}(\Gamma E(a)), \qquad t_a = x_a \oplus x_{a+1} \oplus x_{a+2}.$$

Over the full ring vector $X = (x_0, x_1, \dots, x_{N-1})^T$, the defect sequence is $t = Q_{\mathrm{op}}X$ where $Q_{\mathrm{op}} = I \oplus T \oplus T^2$. Because $N = 2^n$ is a power of two and the Frobenius endomorphism over $\mathbb{F}_2$ is linear, $Q_{\mathrm{op}}^N = (I \oplus T \oplus T^2)^{2^n} = I \oplus T^{2^n} \oplus T^{2^{n+1}} = I \oplus I \oplus I = I$, so $Q_{\mathrm{op}}^{-1} = Q_{\mathrm{op}}^{2^n-1}$ reconstructs $X = Q_{\mathrm{op}}^{2^n-1}t$ from its local three-point defects without an anchor.

Theorem 2.1 (Route-Switch Defect Zeros and Perimeter Line Coverage). *For every width $n \geq 5$, the route-switch defect $t_a = R_n[a] \oplus R_n[a+1] \oplus R_n[a+2]$ vanishes at exactly two addresses, $a = 1$ and $a = 2^n - 4$, and the 2^n perimeter edges $(R_n[a], R_n[a+1])$ lie on exactly $2^n - 4$ distinct projective lines of $\mathrm{PG}(n-1,2)$.*

Proof. For $n \geq 5$, the first four words of R_n are $R_n[0] = 0$, $R_n[1] = 1$, $R_n[2] = 2$, $R_n[3] = 3$, because every $x \in \{1,2,3\}$ has $J_n(x) \geq 2^{n-2} \geq 8 > x$. Thus at $a = 1$: $t_1 = R_n[1] \oplus R_n[2] \oplus R_n[3] = 1 \oplus 2 \oplus 3 = 0$.By the facing law $R_n[N-1-k] = P_n(R_n[k])$ and the linearity of J_n on the non-palindromes $\{1,2,3\}$, the last four words of R_n are $R_n[N-4] = J_n(3) = 3 \cdot 2^{n-2}$, $R_n[N-3] = J_n(2) = 2^{n-2}$, $R_n[N-2] = J_n(1) = 2^{n-1}$, and $R_n[N-1] = P_n(0) = u$. Therefore, at $a = N-4 = 2^n - 4$:

$$t_{N-4} = R_n[N-4] \oplus R_n[N-3] \oplus R_n[N-2] = J_n(3 \oplus 2 \oplus 1) = J_n(0) = 0.$$

Each of the two edges $(R_n[0], R_n[1]) = (0,1)$ and $(R_n[N-1], R_n[0]) = (u, 0)$ contains 0 and therefore defines a zero-gap boundary triad rather than an interior projective line. At $a = 1$, consecutive edges $(1,2)$ and $(2,3)$ share the projective line $\{1,2,3\}$ because $t_1 = 0$; at $a = N-4$, consecutive edges $(J_n(3), J_n(2))$ and $(J_n(2), J_n(1))$ share the mirrored line $\{2^{n-2}, 2^{n-1}, 3 \cdot 2^{n-2}\}$ because $t_{N-4} = 0$. Subtracting the 2 zero-touching edges and the 2 duplicate line visits from $N = 2^n$ leaves $2^n - 4$ distinct projective lines. The argument establishes the two zeros and the two shared lines; that no other t_a vanishes, and that the remaining edges lie on pairwise distinct lines, is verified by direct computation for $5 \leq n \leq 10$. (At the small widths $n = 2,3,4$, boundary overlaps yield zeros at $[1]$ with 1 line for $n = 2$, $[3]$ with 5 lines for $n = 3$, and $[1,3,12]$ with 11 lines for $n = 4$.) ■

2.4 The Boolean Möbius Transform and Local Difference Stack Equivalence

To connect circular transport along the ring's address axis with word-internal operations along the shape axis, decompose R_n into n address-indexed bit planes $x^{(r)} \in \mathbb{F}_2^N$ ($r \in \{0, \dots, n-1\}$, $N = 2^n$) defined by $x_k^{(r)} = \mathrm{bit}_r(R_n[k])$, and let $D_k = R_n[k] \oplus R_n[(k+1) \bmod N]$ denote the adjacent-address XOR instruction stream.

The Boolean Möbius (subset-XOR) transform $M: \mathbb{F}_2^N \to \mathbb{F}_2^N$ is defined on any address tape $x = (x_0, \dots, x_{N-1})^T$ by:

$$a_i = (Mx)_i = \oplus_{k \subseteq i} x_k = \oplus_{k=0}^{N-1} [(k \,\&\sim i) = 0] x_k, \qquad i \in \{0,1, \dots, N-1\}.$$

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Theorem 2.2 (Boolean Möbius Conjugacy and Shift Identity). *On $\mathbb{F}_2^N$ with $N = 2^n$, the Boolean Möbius transform M is self-inverse ($M^2 = I$) and conjugates the cyclic shift operator $(Rx)_k = x_{(k+1) \bmod N}$ into the identity plus the nilpotent superdiagonal shift $(Sa)_i = a_{i+1}$ (for $0 \le i < N-1$) and $(Sa)_{N-1} = 0$:*

$$MRM = I \oplus S, \qquad MD^{(r)} = M(I \oplus R)x^{(r)} = SMx^{(r)} = Sa^{(r)}.$$

Proof. By Lucas' theorem, the binomial coefficient $\binom{i}{k}$ is odd ($\binom{i}{k} \equiv 1 \pmod 2$) if and only if the binary expansion of k is a submask of i ($k \subseteq i$). Consequently, the i-th Möbius coefficient $(Mx)_i$ is identically the i-th forward difference of the tape x evaluated at address $k = 0$:

$a_i = (Mx)_i = \oplus_{k=0}^{i} \left(\binom{i}{k} \bmod 2\right) x_k = ((I \oplus R)^i x)_0$.Applying M to the adjacent-address XOR stream $D^{(r)} = (I \oplus R)x^{(r)}$ therefore yields, for every $0 \le i < N-1$:$(MD^{(r)})_i = ((I \oplus R)^i (I \oplus R)x^{(r)})_0 = ((I \oplus R)^{i+1} x^{(r)})_0 = a_{i+1}^{(r)} = (Sa^{(r)})_i$,

while at the top index $i = N-1$, $(MD^{(r)})_{N-1} = ((I \oplus R)^N x^{(r)})_0 = ((I \oplus R^{2^n})x^{(r)})_0 = 0 = (Sa^{(r)})_{N-1}$. Because M acts linearly along the address index k and word-internal shape operations (such as adjacent-bit XOR $(x_k^{(r)} \oplus x_k^{(r+1)})$) act linearly along the bit index r, the two operators act on orthogonal tensor factors of $\mathbb{F}_2^N \otimes \mathbb{F}_2^n$ and commute identically. Executing this transform across all 77 bit planes for $n = 2$ through $n = 12$ confirms $MD^{(r)} = Sa^{(r)}$ with zero mismatches. ■

3. Single-Reader Retained Continuation, the Carrier Subspace Theorem, and the Price of Motion

3.1 From Fixed-Mold Orbits to the Writable Field Engine

On a fixed mold $M = R_n$, a carried difference Δ updates without writing via $U_B(m, \Delta) = (m \oplus \Delta, B(\Delta))$. When $B = P_n$, the involution $P_n^2 = I$ forces $(m \oplus \Delta) \oplus P_n(\Delta) \oplus \Delta \oplus P_n(\Delta) = m$, so every orbit closes in 4 steps. On the residue pair square $W_r(i,j) = (i \ll r) \mid J_r(j)$, the innermost diamond is this 4-step orbit seeded at $m = m^*, \Delta = e_R$, where every off-diagonal 2×2 cell exhibits an address-step slip $|\mathrm{gap}_{S-1} - \mathrm{gap}_{S+1}| \in \{0,1\}$. Because the mold never changes, fixed-mold continuation leaves no record in the ring.

To retain history, endow R_n with a live writable value field $V: \{0, \ldots, 2^n - 1\} \to \mathcal{W}_n$ initialized at the vacuum mold $V = M$. A reader at address a with carry w reads the effective value $e = V[a] \oplus w$, transitions to $b = \mathrm{READ}_n(e)$, updates its carry to $w' = P_n(e \oplus M[a])$, and writes $V'[a] = V[a] \oplus w'$. Defining the reader's word $p = M[a] \in \mathcal{W}_n$ and the sparse field offset $D[q] = V[\mathrm{READ}_n(q)] \oplus q \in \mathcal{W}_n$ eliminates the ring address ordering from the dynamical step:

$$x = D[p] \oplus w, \qquad p' = p \oplus x, \qquad w' = P_n(x), \qquad D'[q] = \begin{cases} D[p] \oplus w', & \text{if } q = p, \\ D[q], & \text{if } q \neq p. \end{cases}$$

This single-reader step $F(p, w, D) = (p', w', D')$ is inverted in $O(1)$ field operations by recovering $x = P_n(w')$, $p = p' \oplus x$, $D[p] = D'[p] \oplus w'$, and $w = x \oplus D[p]$.

3.2 The Carrier Subspace Theorem and the Three Universal Closure Classes

Theorem 3.1 (Carrier Subspace Confinement and Trichotomy). *For any width $n \ge 2$, let a reader begin at word $p_0 \in \mathcal{W}_n$ with carry $w_0 \in \mathcal{W}_n$ on an empty field $D_0 = 0$. For all ticks $t \ge 0$, the reader position $p(t)$ remains confined to the affine coset $p_0 \oplus K(w_0)$, and every carry $w(t)$ and field offset $D_t[q]$ remains inside the $\mathbb{F}_2$-linear subspace:*

$$K(w_0) = \mathrm{span}_{\mathbb{F}_2}\{u,\, w_0,\, J_n(w_0)\} \subset \mathcal{W}_n.$$

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The subspace $K(w_0)$ has dimension 1, 2, or 3 according to how w_0 meets its mirror $J_n(w_0)$, partitioning all 2^n initial carries into exactly three dynamically invariant closure classes:

1. **Class R_1 ($\dim K = 1$, $|K| = 2$, Period 6):** *When $w_0 \in \{0, u\}$, $K(w_0) = \{0, u\} \cong R_1$. The trajectory closes in 6 ticks, visiting the vacuum $D_t = 0$ on 4 of 6 ticks with mean kept field cells $\bar{k} = 1/3 \approx 0.333333$.*
2. **Class R_2 ($\dim K = 2$, $|K| = 4$, Period 132):** *When $w_0 \notin \{0, u\}$ and $J_n(w_0) \in \{w_0, w_0 \oplus u\}$ (meaning P_n acts as C_n on all of $K(w_0)$), $K(w_0) = \{0, u, w_0, u \oplus w_0\} \cong R_2$. There are $2^{\lfloor (n+2)/2 \rfloor} - 2$ such carries at width n. The trajectory closes in 1 single lap of 132 ticks, visiting the vacuum only at $t = 132$ with mean kept cells $\bar{k} = 83/33 \approx 2.515152$.*
3. **Class R_3 ($\dim K = 3$, $|K| = 8$, Period 43,155,322):** *When $J_n(w_0) \notin \{w_0, w_0 \oplus u\}$ (meaning $P_n(w_0) \neq C_n(w_0)$), the vectors $\{u, w_0, J_n(w_0)\}$ are linearly independent over $\mathbb{F}_2$ and span an 8-word carrier $K(w_0) \cong R_3$. There are $2^n - 2^{\lfloor (n+2)/2 \rfloor}$ such carries (first appearing at the Fano width $n = 3$). In the basis $(e_R, e_L, u) \mapsto (w_0, J_n(w_0), u)$, the dynamics are isomorphic at every width $n \geq 3$ and close in two laps of total period $T = 43{,}155{,}322 = 2 \cdot 7 \cdot 3{,}082{,}523$ ticks, with mean kept cells $\bar{k} = 298{,}630{,}782/43{,}155{,}322 \approx 6.919906$.*

Proof. Because J_n is a linear involution with $J_n(u) = u$, the subspace $K(w_0) = \mathrm{span}_{\mathbb{F}_2}\{u, w_0, J_n(w_0)\}$ contains u and is invariant under J_n. For any $x \in K(w_0)$, $P_n(x)$ is either $J_n(x) \in K(w_0)$ or $x \oplus u \in K(w_0)$, so $K(w_0)$ is closed under P_n. By induction on t, $x(t) = D_t[p(t)] \oplus w(t) \in K(w_0)$, $w(t+1) = P_n(x(t)) \in K(w_0)$, $D_{t+1}[p(t)] \in K(w_0)$, and $p(t+1) \in p_0 \oplus K(w_0)$. On an 8-word carrier $K(w_0)$ with basis $b_0 = w_0$, $b_1 = J_n(w_0)$, $b_2 = u$, any vector $x = x_0 b_0 \oplus x_1 b_1 \oplus x_2 b_2 \in K(w_0)$ has $J_n(x) = x_0 b_1 \oplus x_1 b_0 \oplus x_2 b_2$, so $x \in \mathrm{Pal}_n$ if and only if $x_0 = x_1$, i.e., $\ell_0(x) = x_0 \oplus x_1 = 0$. Thus $P_n|_{K(w_0)}$ depends exclusively on the 3-bit coordinates (x_0, x_1, x_2), rendering the trajectory identical across all widths $n \geq 3$. ■

3.3 The Two-Lap Half-Return Involution

Over the 43,155,322-tick period of the R_3 closure seeded at $(p_0, w_0, D_0) = (0, e_R, 0)$, the field D_t returns to the vacuum $D_t = 0$ on exactly two ticks: the full return at $t = 43{,}155{,}322$ and the exact midpoint $t_{1/2} = 21{,}577{,}661$. At $t_{1/2} = 21{,}577{,}661$, the field is completely empty ($D_{t_{1/2}} = 0$), and the reader stands at the complemented state: $p(21{,}577{,}661) = u$, $w(21{,}577{,}661) = u \oplus e_R$, $D_{21{,}577{,}661} = 0$.Define the linear involution $A_0 : K(w_0) \to K(w_0)$ by $A_0(x) = x \oplus \ell_0(x)u$ (which maps $e_R \mapsto u \oplus e_R$, $e_L \mapsto u \oplus e_L$, and $u \mapsto u$). Extending A_0 with carrier translation by u defines the state-space involution $\sigma(p, w, D) = (A_0(p) \oplus u, A_0(w), D^\sigma)$ where $D^\sigma[A_0(q) \oplus u] = A_0(D[q])$. Because A_0 commutes with P_8, σ commutes identically with F ($\sigma \circ F = F \circ \sigma$ and $\sigma^2 = I$), forcing:

$$F^{21{,}577{,}661}(S_0) = \sigma(S_0), \qquad F^{43{,}155{,}322}(S_0) = \sigma^2(S_0) = S_0.$$

If the palindrome complement branch of P_n is disabled so that the partner is pure bit-reversal J_n, the R_1, R_2, R_3 carries close in 1, 6, and $518 = 2 \cdot 7 \cdot 37$ ticks respectively, and at the half-lap $t = 259$ the reader returns mirrored at $(0, e_L, D = 0)$ under $(J, h = 0)$ rather than complemented.

3.4 Lazy-Frame Transport, Handoffs, and the Gradient Price of Motion

To relocate a closure across the 2^{n-3} cosets of L in $\mathcal{W}_n$, the global transport operator H_h shifts the reader and the field offsets together while leaving the mold $M = R_n$ fixed:

$$H_h(p, w, D) = (p \oplus h,\ w,\ q \mapsto D[q \oplus h]).$$

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Because $D'[p \oplus h] = D[p]$ under H_h, transport satisfies $H_h^2 = I$, $H_a H_b = H_{a \oplus b}$, and commutes with forward and inverse execution: $FH_h = H_h F$ and $F^{-1}H_h = H_h F^{-1}$ (verified across 30,844 deterministic checks in the v3 engine). Introducing the reader-relative field $E[r] = D[p \oplus r]$ and lazy backing store $E[r] = V[r \oplus s]$ with internal frame word $s \in \mathcal{W}_n$ reduces both F and H_h to $O(1)$ native dictionary operations independent of field occupancy: F executes $x = V[s] \oplus w$, $V'[s] = V[s] \oplus P_n(x)$, $s' = s \oplus x$, $p' = p \oplus x$, $w' = P_n(x)$, while H_h updates only $p' = p \oplus h$ with zero field key rewrites.

In the absolute field representation, moving a closure by $h \notin K(w_0)$ shifts its support to a disjoint coset, changing $C_{\text{cell}}(D; h) = \sum_{q \in \mathcal{W}_n} [\, D[q] \neq D[q \oplus h]] = 2k$ field cells, where $k = |\{q : D[q] \neq 0\}|$ is the number of kept cells. Averaged over each closure's period, the gradient move price is $2\bar{k} = 13.839812$ cells for R_3 (where $k \geq 7$ on 70.62% of ticks, and $C_{\text{cell}} = 0$ only at the two vacuum ticks $t = 21{,}577{,}661$ and $t = 43{,}155{,}322$), 5.030303 cells for R_2, and 0.666667 cells for R_1. If the reader is instead kicked alone at tick 4,096 ($p \leftarrow p \oplus o_R$, where $p = 0, w = u, k = 7$), it abandons its 7 kept cells as a static record and enters a 2-word, 6-step R_1 cycle in the new coset; if the carry alone is kicked ($w \leftarrow w \oplus o_R$), the kick interacts with the 7 kept cells to open the 32-word two-shell subspace $W = \text{span}_{\mathbb{F}_2}\{u, e_L, e_R, o_L, o_R\} \cong \text{PG}(4,2)$.

4. Simultaneous Multi-Reader Dynamics and the Four-Involution Factorization Theorem

4.1 The Simultaneous Multi-Reader Law and the Lazy-Frame Lift

When $m \geq 2$ readers $\boldsymbol{r} = ((p_1, w_1), \ldots, (p_m, w_m))$ share a single writable field D, sequential execution $F_1 \circ F_2$ vs. $F_2 \circ F_1$ fails to commute whenever $p_1 = p_2$. The canonical endogenous multi-reader law $F^{(m)}(\boldsymbol{r}, D) = (\boldsymbol{r}', D')$ eliminates ordering ambiguity by requiring all m readers to read the existing field D simultaneously before XOR-combining their writes:

$$x_i = D[p_i] \oplus w_i, \qquad p_i' = p_i \oplus x_i, \qquad w_i' = P_n(x_i), \qquad D'[q] = D[q] \oplus \oplus_{i:p_i=q} w_i'.$$

In the v3 lazy-frame architecture, choosing reader 0 as the coordinate origin sets $r_i = p_i \oplus p_0$ ($r_0 = 0$) and $E[r] = D[p_0 \oplus r] = V[r \oplus s]$. All m readers execute through the shared backing dictionary V without absolute-field copying:

$$x_i = V[r_i \oplus s] \oplus w_i, \qquad w_i' = P_n(x_i), \qquad V'[r_i \oplus s] = V[r_i \oplus s] \oplus \oplus_{j:r_j=r_i} w_j',$$

$$r_i' = r_i \oplus x_i \oplus x_0, \qquad s' = s \oplus x_0, \qquad p_0' = p_0 \oplus x_0.$$

Global carrier transport H_h modifies only the placement origin $p_0 \leftarrow p_0 \oplus h$, leaving $(s, \{r_i\}, \{w_i\}, V)$ literally unchanged. Independent testing across 6,000 randomized multi-reader states and global transports confirms exact agreement between the lazy-frame recurrence and the absolute shared-field engine.

4.2 The Four-Involution Factorization Theorem and Universal Shell-Lamp Conservation

Theorem 4.1 (Four-Involution Factorization and Collision-Tolerant Inverse). *On the m-reader state space $\mathcal{S}_{n,m} = (\mathcal{W}_n \times \mathcal{W}_n)^m \times \mathcal{W}_n^{2^n}$, define four atomic operators $\mathcal{R}, \Pi, \mathcal{W}, \mathcal{M} : \mathcal{S}_{n,m} \to \mathcal{S}_{n,m}$:*

$$\mathcal{R}(((p_i, w_i))_{i=1}^m, D) = (((p_i, D[p_i] \oplus w_i))_{i=1}^m, D), \qquad \Pi(((p_i, w_i))_{i=1}^m, D) = (((p_i, P_n(w_i)))_{i=1}^m, D),$$

$$\mathcal{W}(((p_i, w_i))_{i=1}^m, D) = \big(((p_i, w_i))_{i=1}^m,\ q \mapsto D[q] \oplus \oplus_{i:p_i=q} w_i\big), \qquad \mathcal{M}(((p_i, w_i))_{i=1}^m, D) = (((p_i \oplus P_n(w_i), w_i))_{i=1}^m, D).$$

Then $\mathcal{R}^2 = \Pi^2 = \mathcal{W}^2 = \mathcal{M}^2 = I$, and the simultaneous multi-reader step $F^{(m)}$ and its exact inverse $(F^{(m)})^{-1}$ factor as:

$$F^{(m)} = \mathcal{M} \circ \mathcal{W} \circ \Pi \circ \mathcal{R}, \qquad (F^{(m)})^{-1} = \mathcal{R} \circ \Pi \circ \mathcal{W} \circ \mathcal{M}.$$

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Proof. Each of $\mathcal{R}, \Pi, \mathcal{W}, \mathcal{M}$ modifies only one coordinate using an operation of order 2 (XOR addition or $P_n^2 = I$) while holding the remaining coordinates fixed, so each is an exact involution. Applying $\mathcal{R}, \Pi, \mathcal{W}, \mathcal{M}$ sequentially to $(((p_i, w_i)), D)$ yields:

$$(((p_i, w_i)), D) \xrightarrow{\mathcal{R}} (((p_i, x_i)), D) \xrightarrow{\Pi} (((p_i, w_i')), D) \xrightarrow{\mathcal{W}} (((p_i, w_i')), D') \xrightarrow{\mathcal{M}} (((p_i \oplus P_n(w_i'), w_i')), D') = F^{(m)}(((p_i, w_i)), D),$$

since $P_n(w_i') = P_n(P_n(x_i)) = x_i$. Because each factor is an involution, $(F^{(m)})^{-1} = \mathcal{R}^{-1} \circ \Pi^{-1} \circ \mathcal{W}^{-1} \circ \mathcal{M}^{-1} = \mathcal{R} \circ \Pi \circ \mathcal{W} \circ \mathcal{M}$, which explicitly computes $x_i = P_n(w_i')$, $p_i = p_i' \oplus x_i$, $D[q] = D'[q] \oplus \oplus_{i:p_i=q} w_i'$, and $w_i = x_i \oplus D[p_i]$. Verified with zero errors across 10,000 randomized 32-word states and 2,000 5-reader collision states. ■

For each mirror-paired shell $k \in \{0, \dots, \lfloor n/2 \rfloor - 1\}$, define the 1-bit shell-lamp functional $\ell_k(x) = x_k \oplus x_{n-1-k} \in \mathbb{F}_2$. Because J_n swaps $x_k \leftrightarrow x_{n-1-k}$ and C_n flips both bits ($1 \oplus x_k \oplus 1 \oplus x_{n-1-k} = x_k \oplus x_{n-1-k}$), ℓ_k is invariant under the native partner: $\ell_k(P_n(x)) = \ell_k(x)$ for all $x \in \mathcal{W}_n$. Consequently, under $F^{(m)}$, the total shell-lamp charge:

$$\mathcal{L}_k = \oplus_{i=1}^m \ell_k(p_i) \oplus \oplus_{q \in \mathcal{W}_n} \ell_k(D[q])$$

updates by $\mathcal{L}_k' = \oplus_{i=1}^m \ell_k(p_i \oplus x_i) \oplus \oplus_q \ell_k(D[q]) \oplus \oplus_{i=1}^m \ell_k(P_n(x_i)) = \mathcal{L}_k \oplus \oplus_{i=1}^m (\ell_k(x_i) \oplus \ell_k(x_i)) = \mathcal{L}_k$, proving exact conservation of $\mathcal{L}_k$ for any number of readers and collisions.

4.3 The Three-Bit Affine Conservation Law and Vacuum Separation Theorem

On any 8-word single-shell carrier $L = \mathrm{span}_{\mathbb{F}_2}\{e_R, e_L, u\} \cong \mathbb{F}_2^3$ with coordinates $e_R = 1, e_L = 2, s = 3, u = 4$ and center-shell lamp $\ell_0(x) = x_0 \oplus x_1$, a word $x \in L$ is a palindrome if and only if $\ell_0(x) = 0$. When $\ell_0(x) = 0$, $P_8(x) = x \oplus u$; when $\ell_0(x) = 1$, $P_8(x) = J_8(x) = x \oplus s$. Combining both branches expresses P_8 as an exact $\mathbb{F}_2$-affine transformation:

$$P_8(x) = A(x) \oplus u, \qquad A(x) = x \oplus \ell_0(x)(u \oplus s) = x \oplus 7 \cdot \ell_0(x), \qquad A^2 = I.$$

Theorem 4.2 (Three-Bit Conserved Charge and Vacuum Separation Law). *For m readers executing on the 8-word carrier L, let $B_D = \oplus_{q \in L} D[q]$ and $r = \oplus_{i=1}^m p_i$. Under one step of $F^{(m)}$, the 3-bit vector $\mathcal{Q}_m = B_D \oplus A(r)$ transforms as $\mathcal{Q}_m' = \mathcal{Q}_m \oplus (m \bmod 2)u$. In particular, for two readers ($m = 2$), the 3-bit vector:*

$$\mathcal{Q} = B_D \oplus A(p_A \oplus p_B) \in L$$

is an exact invariant of execution ($\mathcal{Q}' = \mathcal{Q}$). Whenever the field returns to the vacuum ($D_t = 0$) from an initially empty field ($D_0 = 0$), the two readers' XOR separation is identically preserved:

$$D_t = 0 \Longrightarrow p_A(t) \oplus p_B(t) = p_A(0) \oplus p_B(0).$$

Proof. Because A is linear over $\mathbb{F}_2$ and $P_8(x_i) = A(x_i) \oplus u$, the total field XOR $B_D = \oplus_{q \in L} D[q]$ updates under $m = 2$ simultaneous writes by:

$$B_D' = B_D \oplus P_8(x_A) \oplus P_8(x_B) = B_D \oplus (A(x_A) \oplus u) \oplus (A(x_B) \oplus u) = B_D \oplus A(x_A \oplus x_B).$$

Simultaneously, the relative reader separation $r = p_A \oplus p_B$ updates by $r' = r \oplus x_A \oplus x_B$, so $A(r') = A(r) \oplus A(x_A \oplus x_B)$. Adding the two equations gives $\mathcal{Q}' = B_D' \oplus A(r') = B_D \oplus A(r) = \mathcal{Q}$. At any tick t where $D_t = 0$, $B_{D_t} = 0$, so $A(p_A(t) \oplus p_B(t)) = \mathcal{Q} = A(p_A(0) \oplus p_B(0))$. Because $A^2 = I$ is invertible, applying A to both sides yields $p_A(t) \oplus p_B(t) = p_A(0) \oplus p_B(0)$. ■

5. The Vacuum Monodromy Compiler and Exact Resolution of All 64 Same-Carrier Two-Reader Seeds

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5.1 The 64-Element Symmetry Group of the Two-Reader Machine

On the 8-word carrier $L \cong \mathbb{F}_2^3$, an invertible linear map $g \in \mathrm{GL}_3(\mathbb{F}_2)$ commutes with the affine partner $P_8(x) = x \oplus \ell_0(x)(u \oplus s) \oplus u$ if and only if $g(u) = u$, $g(s) = s$, and $\ell_0(g(x)) = \ell_0(x)$ for all $x \in L$. Exhaustive enumeration of all $|\mathrm{GL}_3(\mathbb{F}_2)| = 168$ invertible matrices confirms that the centralizer of P_8 in $\mathrm{GL}_3(\mathbb{F}_2)$ is the Klein four-group $C_{\mathrm{GL}_3(\mathbb{F}_2)}(P_8) = \{I, J, A_0, JA_0\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, where in the basis $(e_R, e_L, u) = (1,2,4)$:

$I = (1,2,4)$, $J = (2,1,4)$, $A_0 = (5,6,4)$, $JA_0 = (6,5,4)$.Combining $g \in \{I, J, A_0, JA_0\}$ with the 8 carrier translations H_h ($h \in L$) and reader label exchange sw $\in \{0,1\}$ yields the 64-element equivariance group $\mathcal{G}_{64} = (\mathbb{Z}_2 \times \mathbb{Z}_2) \ltimes \mathbb{F}_2^3 \times \mathbb{Z}_2$, acting on two-reader states $S = (((p_A, w_A), (p_B, w_B)), D)$ by:$\gamma_{(g,h,\mathrm{sw})}(S) = \Big(\big((g(p_{\pi(A)}) \oplus h,\ g(w_{\pi(A)})),\ (g(p_{\pi(B)}) \oplus h,\ g(w_{\pi(B)}))\big),\ q \mapsto g(D[g(q \oplus h)])\Big)$,

where $(\pi(A), \pi(B)) = (B, A)$ if sw $= 1$ and (A, B) if sw $= 0$, and every $\gamma \in \mathcal{G}_{64}$ commutes identically with execution: $F^{(2)} \circ \gamma = \gamma \circ F^{(2)}$.

5.2 Complete Classification of the 20 Short Cycles and Midpoint Structure of One-Lap Pairs

Fixing reader A at $(p_A(0), w_A(0)) = (0, e_R)$ on an empty field $D_0 = 0$ defines 64 canonical two-reader seeds indexed by $\sigma = (p_B(0), w_B(0)) \in L \times L$. Within the 25,000-tick horizon, exactly 20 seeds complete their full period: 1 silent pair, 8 one-lap pairs, and 11 two-lap pairs.

1. **The Silent Identical Pair ($T = 4$, 1 seed):** When $B = (0, e_R)$, both readers start at the same word with the same carry. At every tick t, $x_A(t) = x_B(t)$, so their simultaneous writes $P_8(x_A) \oplus P_8(x_B) = 0$ cancel identically, leaving $D_t = 0$ forever while both readers trace the 4-step fixed-mold diamond $0 \to e_R \to s \to e_L \to 0$.
2. **The 11 Short Two-Lap Cycles ($T \in \{264,288,1612,1950,4956,6246,9750,13466,22750\}$):** Every two-lap cycle empties its field ($D_{T/2} = 0$) at the exact half-period $t = T/2$, where $F^{T/2}(S_0) = \gamma_{1/2}(S_0)$ for a fixed-point-free involution $\gamma_{1/2} = (g, h, \mathrm{sw}) \in \mathcal{G}_{64}$ satisfying $g(h) = h$. Notably, the three seeds $B \in \{(e_R, e_R), (u \oplus e_R, e_R), (u \oplus e_L, e_R)\}$ belong to a single 22,750-tick macro-trajectory with 6 vacuum visits; for $B = (e_R, e_R)$, the vacuum visits occur at ticks 883 (landing on a $\mathcal{G}_{64}$-image of $(u \oplus e_L, e_R)$), 4,081 (landing on $(u \oplus e_R, e_R)$), $11{,}375 = T/2$ (landing on its own half-lap image under $(I, u, 0)$), 12,258, 15,456, and 22,750.
3. **The 8 One-Lap Cycles ($T \in \{26,132,258,1554,4936\}$) and Midpoint Symmetry:** None of the 8 one-lap cycles visits the vacuum before $t = T$. However, inspecting the midpoint state $S_{T/2}$ at $t = T/2$ reveals why they close in a single lap: at $t = T/2$, every one-lap cycle reaches a populated field $D_{T/2} \neq 0$ whose total field sum vanishes ($B_D(T/2) = \oplus_q D_{T/2}[q] = 0$), forcing $p_A(T/2) \oplus p_B(T/2) = p_B(0)$ even though $D_{T/2} \neq 0$. Moreover, the three equal-period one-lap pairs—$B = (0, e_L)$ and (s, e_L) at $T = 26$; $B = (0, u \oplus e_L)$ and $(u \oplus s, u \oplus e_L)$ at $T = 132$; $B = (0, u \oplus e_R)$ and $(u, u \oplus e_R)$ at $T = 258$—satisfy $p_B^{(2)}(0) = w_A(0) \oplus w_B(0) = e_R \oplus w_B(0)$, so that at $t = 1$ the first seed steps to $p_A(1) \neq p_B(1)$ while the second seed collides at $p_A(1) = p_B(1) = e_R$ with distinct carries $w_A(1) = e_L$ and $w_B(1) = P_8(w_B(0)) \in \{e_R, u \oplus e_R, u \oplus e_L\}$.

5.3 The Vacuum Monodromy Compiler and Exact Closure of the 28 Open Seeds

In the 300,000,000-tick C census of the 64 same-carrier seeds, 36 seeds closed (the 20 short seeds plus 16 long closures with periods between 6,139,402 and 291,539,620), while 28 seeds remained open beyond

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300,000,000 ticks. Yet every one of those 28 open seeds recorded between 13 and 18 intermediate vacuum returns ($D_t = 0$), and the C census printed the exact tick $t_1(\sigma)$ and reader state $S_{t_1}(\sigma) = ((p_A(t_1), w_A(t_1)), (p_B(t_1), w_B(t_1)), D = 0)$ of the first vacuum return for all 64 seeds.

Theorem 5.1 (Vacuum Monodromy Compiler and Complete 64-Seed Closure). Let $\Sigma = L \times L$ denote the 64 canonical empty-field seeds $S_0(\sigma) = (((0, e_R), \sigma), D = 0)$. Every seed has a reader in the R_3 carry class ($w_A(0) = e_R$), and each of the 64 recorded first vacuum return states $F^{t_1(\sigma)}(S_0(\sigma))$ also contains, as the census records show, at least one reader with carry in $R_3 = \{e_R, e_L, u \oplus e_R, u \oplus e_L\}$, on which $C_{\mathrm{GL}_3(\mathbb{F}_2)}(P_8) = \{I, J, A_0, JA_0\}$ acts transitively. Consequently, for every $\sigma \in \Sigma$, there exists a transition symmetry $\gamma(\sigma) \in \mathcal{G}_{64}$ and a next canonical seed $\pi(\sigma) \in \Sigma$ such that:

$$F^{t_1(\sigma)}(S_0(\sigma)) = \gamma(\sigma)(S_0(\pi(\sigma))).$$

By $\mathcal{G}_{64}$-equivariance of $F^{(2)}$, the k-th vacuum return of $S_0(\sigma)$ occurs at tick $T_k(\sigma) = \sum_{j=0}^{k-1} t_1(\pi^j(\sigma))$ in the exact state:

$$F^{T_k(\sigma)}(S_0(\sigma)) = (\gamma(\sigma) \circ \gamma(\pi(\sigma)) \circ \ldots \circ \gamma(\pi^{k-1}(\sigma)))(S_0(\pi^k(\sigma))).$$

Evaluating this monodromy cocycle across all 64 seeds resolves the complete phase space of the two-reader 8-word carrier: all 64 seeds close in finite time, and the 28 seeds open past 300,000,000 ticks partition into exactly two closed macro-orbits:

1. **The 10-Seed Macro-Orbit ($T = 310{,}826{,}368$, 16 Vacuum Returns, Two Laps):** Containing the 10 seeds (e_R, e_L), (e_L, e_L), (e_L, s), (e_L, u), $(e_L, u \oplus e_L)$, $(e_L, u \oplus s)$, $(u \oplus e_R, 0)$, $(u \oplus e_R, u)$, $(u \oplus e_R, u \oplus e_L)$, and $(u \oplus e_R, u \oplus s)$. Every seed in this family has exact period $T = 310{,}826{,}368$ ticks, visits the vacuum 16 times per period (13 to 15 times before 300,000,000 ticks), and reaches its half-period vacuum at $t_{1/2} = 155{,}413{,}184$ under the pure carrier translation $(I, u \oplus e_L, 0)$, $(I, u \oplus e_R, 0)$, or $(I, e_L, 0)$.
2. **The 18-Seed Macro-Orbit ($T = 515{,}455{,}952$, 28 Vacuum Returns, Two Laps):** *Containing the 18 seeds $(e_R, 0)$, (e_R, s), (e_R, u), $(e_R, u \oplus e_R)$, $(e_R, u \oplus e_L)$, $(e_R, u \oplus s)$, $(e_L, 0)$, $(e_L, u \oplus e_R)$, $(u \oplus e_R, e_L)$, $(u \oplus e_R, s)$, $(u \oplus e_R, u \oplus e_R)$, $(u \oplus e_L, 0)$, $(u \oplus e_L, e_L)$, $(u \oplus e_L, s)$, $(u \oplus e_L, u)$, $(u \oplus e_L, u \oplus e_R)$, $(u \oplus e_L, u \oplus e_L)$, and $(u \oplus e_L, u \oplus s)$. Every seed in this family has exact period $T = 515{,}455{,}952$ ticks, visits the vacuum 28 times per period (14 to 18 times before 300,000,000 ticks), and reaches its half-period vacuum at $t_{1/2} = 257{,}727{,}976$ under the pure reader-exchange involution $(I, h = 0, \mathrm{sw} = 1)$:*

$$F^{257{,}727{,}976}((p_A, w_A), (p_B, w_B), D = 0) = ((p_B, w_B), (p_A, w_A), D = 0).$$

Table 2 presents the complete verified census of all 64 canonical seeds on the 8-word carrier, grouped into their 26 orbit families, one for each of 26 distinct period values.

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Orbit Family / Period T	Seed Count	Representative Seed(s) B=(pB,wB)	Initial Charge (ℓo(pB),Qo)	Total Vacuum Visits	Half-Period Tick T/2	Half-Period Involution γ1/2 =(g,h,sw)	Closure Class
4	1	(0, eR)	(0, 0)	4	—	—	Silent Identical
26	2	(0, eL), (s, eL)	(0, 0), (0, s)	1	—	—	One Lap
132	2	(0, u+eL), (u+s, u+eL)	(0, 0), (0, u+s)	1	—	—	One Lap
258	2	(0, u+eR), (u, u+eR)	(0, 0), (0, u)	1	—	—	One Lap
264	1	(u+s, eR)	(0, u+s)	2	132	(JA0, 0, 0)	Two Laps
288	1	(s, eR)	(0, s)	2	144	(A0, 0, 0)	Two Laps
1,554	1	(u, eR)	(0, u)	1	—	—	One Lap
1,612	1	(u, u+eL)	(0, u)	2	806	(I, u+s, 0)	Two Laps
1,950	1	(u+s, u+eR)	(0, u+s)	2	975	(I, s, 1)	Two Laps
4,936	1	(u+s, eL)	(0, u+s)	1	—	—	One Lap

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Orbit Family / Period T	Seed Count	Representative Seed(s) B=(pB,wB)	Initial Charge (ℓo(pB),Qo)	Total Vacuum Visits	Half-Period Tick T/2	Half-Period Involution γ1/2 =(g,h,sw)	Closure Class
4,956	1	(u, eL)	(0, u)	2	2,478	(A0, u, 1)	Two Laps
6,246	1	(s, u+eL)	(0, s)	2	3,123	(I, u+s, 1)	Two Laps
9,750	1	(s, u+eR)	(0, s)	2	4,875	(J, 0, 0)	Two Laps
13,466	1	(eL, eR)	(1, u+eR)	2	6,733	(I, u, 0)	Two Laps
22,750	3	(eR, eR), (u+eR, eR), (u+eL, eR)	(1, u+eL/eL/eR)	6	11,375	(I, u, 0) / (I, eR/eL, 1)	Two Laps
6,139,402	1	(s, 0)	(0, s)	2	3,069,701	(I, eR, 1)	Two Laps
46,493,460	2	(u, 0), (u, u)	(0, u)	8	23,246,730	(I, u+s, 0)	Two Laps
80,114,164	1	(0, 0)	(0, 0)	4	40,057,082	(I, s, 0)	Two Laps
87,447,854	2	(0, u), (0, u+s)	(0, 0)	4	43,723,927	(JA0, u/s, 0)	Two Laps
93,073,108	1	(0, s)	(0, 0)	4	46,536,554	(I, u, 0)	Two Laps

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Orbit Family / Period T	Seed Count	Representative Seed(s) B=(pB,wB)	Initial Charge (ℓ0(pB),Q0)	Total Vacuum Visits	Half-Period Tick T/2	Half-Period Involution γ1/2 =(g,h,sw)	Closure Class
96,103,756	2	(u+s, 0), (u+s, s)	(0, u+s)	4	48,051,878	(I, u, 1)	Two Laps
174,783,204	3	(s, s), (s, u), (s, u+s)	(0, s)	6	87,391,602	(I, u+eR/u+eL, 1)	Two Laps
182,271,500	2	(u+s, u), (u+s, u+s)	(0, u+s)	8	91,135,750	(I, u, 0)	Two Laps
291,539,620	2	(u, s), (u, u+s)	(0, u)	8	145,769,810	(I, u+s, 0)	Two Laps
310,826,368	10	(eR, eL), (eL, eL/s/u/u+eL/u+s), (u+eR, 0/u/u+eL/u+s)	(1, *)	16	155,413,184	(I, u+eL/u+eR/eL, 0)	Two Laps
515,455,952	18	(eR, 0/s/u/u+eR/u+eL/u+s), (eL, 0/u+eR), (u+eR, eL/s/u+eR), (u+eL, !=eR)	(1, *)	28	257,727,976	(I, 0, 1) (Pure Swap)	Two Laps

Ordering the 8 initial positions $p_B(0)$ by their center-shell lamp $\ell_0(p_B) = 0$ ($p_B \in \{0, s, u, u \oplus s\}$) versus $\ell_0(p_B) = 1$ ($p_B \in \{e_R, e_L, u \oplus e_R, u \oplus e_L\}$) and ordering the 8 initial carries $w_B(0)$ by R_3 carries ($\{e_R, e_L, u \oplus e_R, u \oplus e_L\}$) versus $R_1 \cup R_2$ carries ($\{0, u, s, u \oplus s\}$) partitions the 8×8 seed matrix in Table 2 into four 4×4 blocks governed by three exact selection rules:

1. **The Non-Closure of Mixed Carry Classes in Short Horizons:** All 20 short cycles ($T \leq 25{,}000$) occur exclusively when $w_B(0) \in R_3 = \{e_R, e_L, u \oplus e_R, u \oplus e_L\}$. Whenever $w_B(0) \in R_1 \cup R_2 = \{0, u, s, u \oplus s\}$, the period exceeds 6,000,000 ticks.
2. **The $\ell_0(p_B) = 0$ Sub-Grid (32 Seeds):** When $\ell_0(p_B(0)) = 0$ ($p_B(0) \in \{0, s, u, u \oplus s\}$), all 16 seeds in the upper-left 4×4 block ($w_B(0) \in R_3$) close within 9,750 ticks (1 silent, 8 one-lap, 7 two-lap), and

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all 16 seeds in the upper-right 4×4 block ($w_B(0) \in R_1 \cup R_2$) close in intermediate two-lap orbits between 6,139,402 and 291,539,620 ticks with 2 to 8 vacuum visits.

3. **The $\ell_0(p_B) = 1$ Sub-Grid (32 Seeds):** When $\ell_0(p_B(0)) = 1$ ($p_B(0) \in \{e_R, e_L, u \oplus e_R, u \oplus e_L\}$), not a single one-lap cycle exists. For $w_B(0) = e_R$, all 4 seeds close in short two-lap cycles ($T = 13{,}466$ with 2 vacuum visits for $p_B = e_L$, and $T = 22{,}750$ with 6 vacuum visits for $p_B \in \{e_R, u \oplus e_R, u \oplus e_L\}$); the remaining 28 seeds across the lower-left and lower-right blocks form the two macro-orbits of periods 310,826,368 (16 vacuum visits) and 515,455,952 (28 vacuum visits).

6. Cross-Shell Quadratic Partner Algebra and the 28-Word Exact Confinement Theorem

6.1 The Quadratic Polynomial Law of the Two-Shell Native Partner

When two readers interact across two mirror-paired shells, their joint carrier is the 32-word, 5-dimensional subspace $W = L \oplus K = \mathrm{span}_{\mathbb{F}_2}\{e_R, e_L, o_R, o_L, u\} \cong \mathbb{F}_2^5$, where $L = \mathrm{span}_{\mathbb{F}_2}\{e_R, e_L, u\}$ is the center-shell carrier and $K = \mathrm{span}_{\mathbb{F}_2}\{o_R, o_L, u\}$ is the outer-shell carrier. In the basis $(e_R, e_L, o_R, o_L, u) = (1,2,4,8,16)$, the two shell-lamp functionals are:

$$\ell_0(x) = x_0 \oplus x_1, \qquad \ell_1(x) = x_2 \oplus x_3,$$

and the two-shell bit-reversal is the linear involution $J_5(x) = x \oplus \ell_0(x)s \oplus \ell_1(x)o_s$, where $s = e_R \oplus e_L = 3$ and $o_s = o_R \oplus o_L = 12$.

Theorem 6.1 (Quadratic Polynomial Representation of P_5 and Breakdown of $\mathcal{Q}$). *On the 32-word two-shell carrier $W \cong \mathbb{F}_2^5$, a word $x \in W$ is a palindrome ($J_5(x) = x$) if and only if both shell lamps vanish ($\ell_0(x) = 0$ and $\ell_1(x) = 0$). Consequently, the native partner $P_5: W \to W$ has the exact quadratic Boolean polynomial representation:*

$$\begin{aligned} P_5(x) &= J_5(x) \oplus (1 \oplus \ell_0(x))(1 \oplus \ell_1(x))u \\ &= x \oplus \ell_0(x)s \oplus \ell_1(x)o_s \oplus (1 \oplus \ell_0(x) \oplus \ell_1(x) \oplus \ell_0(x)\ell_1(x))u. \end{aligned}$$

For two simultaneous readers A and B on W, the single-step change in total field XOR $B_D = \bigoplus_{q \in W} D[q]$ is:

$$\begin{aligned} B_D' \oplus B_D &= P_5(x_A) \oplus P_5(x_B) \\ &= J_5(x_A \oplus x_B) \oplus [\ell_0(x_A \oplus x_B) \oplus \ell_1(x_A \oplus x_B) \oplus \ell_0(x_A)\ell_1(x_A) \oplus \ell_0(x_B)\ell_1(x_B)]u. \end{aligned}$$

Proof. Because $J_5(x) = x \oplus \ell_0(x)s \oplus \ell_1(x)o_s$ and $\{s, o_s\}$ are linearly independent, $J_5(x) = x$ if and only if $\ell_0(x) = \ell_1(x) = 0$, which over $\mathbb{F}_2$ is equivalent to $(1 \oplus \ell_0(x))(1 \oplus \ell_1(x)) = 1$. Multiplying this indicator by u and adding $J_5(x)$ matches $P_5(x)$ on all 32 words. Whereas the linear terms $\ell_0(x_A) \oplus \ell_0(x_B) = \ell_0(x_A \oplus x_B)$ and constant $u \oplus u = 0$ combine into a function of $x_A \oplus x_B$, the degree-2 cross-term $\ell_0(x_A)\ell_1(x_A) \oplus \ell_0(x_B)\ell_1(x_B)$ does not factor through $x_A \oplus x_B$ whenever both shells are active. Thus the 3-bit affine conservation law $\mathcal{Q}$ holds on any single shell (where either $\ell_1 \equiv 0$ or $\ell_0 \equiv 0$, annihilating the quadratic product $\ell_0\ell_1$), and breaks precisely when a trajectory crosses into a second shell. ■

6.2 Shell-Exchange Invariance and the Cross-Shell Conserved Charge $\mathcal{Q}_{\mathrm{shell}}$

In the cross-shell experiments, reader A starts on L with carry $e_R = 1$ and reader B starts on K with carry $o_R = 4$ on an empty field $D_0 = 0$, in two configurations:

1. **Shared Start:** $p_A(0) = 0, w_A(0) = e_R$ and $p_B(0) = 0, w_B(0) = o_R$ (first interaction at tick 5; mean kept cells 23.810 of 32 over 2,000,000,000 ticks; each reader visits exactly 28 of 32 words).

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2. **Offset Start:** $p_A(0) = 0, w_A(0) = e_R$ and $p_B(0) = s, w_B(0) = o_R$ (first interaction at tick 3; mean kept cells 28.569 of 32 over 2,000,000,000 ticks; each reader visits all 32 words).

Why does Shared Start visit 28 words and experience identical placement transitions for A and B (1,209,156,429 placement changes each over 2,000,000,000 ticks), whereas Offset Start visits all 32 words? Define the linear shell-swap involution $S_{\text{shell}}: W \to W$ by swapping the inner and outer shell generators while fixing u:

$$S_{\text{shell}}(e_R) = o_R, \qquad S_{\text{shell}}(o_R) = e_R, \qquad S_{\text{shell}}(e_L) = o_L, \qquad S_{\text{shell}}(o_L) = e_L, \qquad S_{\text{shell}}(u) = u.$$

Because $\ell_0(S_{\text{shell}}(x)) = \ell_1(x)$ and $\ell_1(S_{\text{shell}}(x)) = \ell_0(x)$, the symmetric quadratic factor $(1 \oplus \ell_0(x))(1 \oplus \ell_1(x))$ is invariant under S_{shell}, so S_{shell} commutes with P_5 on all 32 words: $S_{\text{shell}}(P_5(x)) = P_5(S_{\text{shell}}(x))$. Consequently, the combined state-space involution $\Sigma_{\text{shell}} = (S_{\text{shell}}, h = 0, \text{sw} = 1)$ commutes with $F^{(2)}$.

Theorem 6.2 (Shell-Exchange Invariance and Restoration of Cross-Shell Conservation). *In Cross-Shell Shared Start, the initial state $S_0 = (((0, e_R), (0, o_R)), D = 0)$ is a fixed point of Σ_{shell}. Therefore, for all ticks $t \geq 0$:*

$$p_B(t) = S_{\text{shell}}(p_A(t)), \qquad w_B(t) = S_{\text{shell}}(w_A(t)), \qquad D_t[S_{\text{shell}}(q)] = S_{\text{shell}}(D_t[q]).$$

Moreover, let $\delta = I \oplus S_{\text{shell}}: W \to V_{\text{diag}}$ denote the linear diagonal map onto the 2-dimensional subspace $V_{\text{diag}} = \text{span}_{\mathbb{F}_2}\{e_R \oplus o_R, e_L \oplus o_L\} = \{0,5,10,15\}$; its kernel is $\text{Fix}(S_{\text{shell}})$ and $\delta^2 = 0$. Because $\delta(u) = u \oplus u = 0$, applying δ to $P_5(x)$ annihilates the quadratic u-kick for every $x \in W$:

$$\delta(P_5(x)) = J_5(\delta(x)).$$

Consequently, the Shared-Start trajectory satisfies the exact cross-shell conservation law for all $t \geq 0$:

$$Q_{\text{shell}}(t) = B_D(t) \oplus J_5(p_A(t) \oplus p_B(t)) = 0, \qquad B_D(t) = \oplus_{q \in W} D_t[q] \in V_{\text{diag}}.$$

Proof. Because $\Sigma_{\text{shell}} \circ F^{(2)} = F^{(2)} \circ \Sigma_{\text{shell}}$ and $\Sigma_{\text{shell}}(S_0) = S_0$, $F^{(2)t}(S_0)$ is fixed by Σ_{shell} for all $t \geq 0$. Thus $x_B(t) = S_{\text{shell}}(x_A(t))$, so at each tick the total field XOR $B_D(t)$ changes by:

$$B_D(t+1) \oplus B_D(t) = P_5(x_A(t)) \oplus S_{\text{shell}}(P_5(x_A(t))) = \delta(P_5(x_A(t))).$$

Since $P_5(x) = J_5(x) \oplus \phi(x)u$ with $\phi(x) \in \{0,1\}$ and δ is linear with $\delta(u) = 0$ and $\delta \circ J_5 = J_5 \circ \delta$, we have $\delta(P_5(x_A(t))) = J_5(\delta(x_A(t))) = J_5(x_A(t) \oplus x_B(t))$. Meanwhile, $p_A(t) \oplus p_B(t)$ changes by $x_A(t) \oplus x_B(t)$, so $J_5(p_A(t) \oplus p_B(t))$ changes by $J_5(x_A(t) \oplus x_B(t))$. The two increments cancel identically, and since $B_D(0) = 0$ and $p_A(0) \oplus p_B(0) = 0$, $Q_{\text{shell}}(t) = 0$ for all $t \geq 0$. ■

6.3 Proof of the 28-Word Confinement and 128-Invariant Theorem

Theorem 6.2 also explains why Shared Start never visits the 4 words $\{5,10,21,26\} = \{e_R \oplus o_R, e_L \oplus o_L, u \oplus e_R \oplus o_R, u \oplus e_L \oplus o_L\}$. Notice that the 8-word fixed-point subspace $\text{Fix}(S_{\text{shell}}) = \ker(\delta) = \{0,5,10,15,16,21,26,31\}$ splits into two 4-word sets by lamp parity:

$$\text{Fix}_0 = \{q \in \text{Fix}(S_{\text{shell}}): \ell_0(q) = \ell_1(q) = 0\} = \{0,15,16,31\}, \qquad \text{Fix}_1 = \{q \in \text{Fix}(S_{\text{shell}}): \ell_0(q) = \ell_1(q) = 1\} = \{5,10,21,26\}.$$

Theorem 6.3 (28-Word Confinement, Singular Lamp-Matrix Invariant, and 128 Affine Invariants). *For all ticks $t \in [0, \infty)$ of the Cross-Shell Shared-Start trajectory from $A = (0, e_R), B = (0, o_R), D_0 = 0$:*

1. **Cell-Subspace Confinement (128 Affine Invariants):** Each field cell $D_t[q]$ ($q \in W$) remains inside an exact $\mathbb{F}_2$-linear subspace $U_q \subset W: \dim U_q = 0$ ($U_q = \{0\}$) for the 4 cells $q \in \text{Fix}_1 = \{5,10,21,26\}$;

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$\dim U_q = 1$ ($U_q = \{0,15\}$) for $q \in \{15,31\}$; $\dim U_q = 2$ ($U_q = \{0,5,10,15\}$) for $q \in \{0,16\}$; $\dim U_q = 3$ for the 20 cells with $\ell_0(q) \neq \ell_1(q)$ or $q \in \{6,9,22,25\}$; and $\dim U_q = 4$ for $q \in \{3,12,19,28\}$. Across the 20 independent S_{shell}-orbits of cells, the field spans $\sum_{q \le S_{\text{shell}}(q)} \dim U_q = 44$ independent bits out of 160, and the full 180-bit state vector (p_A, w_A, p_B, w_B, D) is confined to a 52-dimensional linear subspace (codimension 128 invariants).

2. **288-State Reader-Carry Closure and Singular Lamp Determinant:** The pair $(p_A(t), w_A(t))$ is permanently confined to a 288-element subset $\Omega \subset (W \backslash \text{Fix}_1) \times (W \backslash \text{Fix}_1)$ on which the 2×2 shell-lamp matrix is singular over $\mathbb{F}_2$ for all $t \geq 0$:

$$\det_{\mathbb{F}_2} \begin{pmatrix} \ell_0(p_A(t)) & \ell_0(w_A(t)) \\ \ell_1(p_A(t)) & \ell_1(w_A(t)) \end{pmatrix} = \ell_0(p_A(t))\ell_1(w_A(t)) \oplus \ell_1(p_A(t))\ell_0(w_A(t)) = 0.$$

In particular, neither reader ever visits or writes to $\text{Fix}_1 = \{5,10,21,26\}$, bounding the reachable carrier of each reader at $32 - 4 = 28$ words for all $t \in [0, \infty)$.

Proof. Because $D_t[S_{\text{shell}}(q)] = S_{\text{shell}}(D_t[q])$, at any tick where reader A stands at $p_A = q$ with carry w_A and reads a field value $v = D_t[q] \in U_q$, the effective value is $x_A = v \oplus w_A$, the next reader-carry pair is $(p'_A, w'_A) = (q \oplus x_A, P_5(x_A))$, and the write increment to cell q is $\Delta D[q] = w'_A \oplus S_{\text{shell}}(w'_A) = \delta(w'_A)$ if $q \in \text{Fix}(S_{\text{shell}})$ (where $p_A = p_B = q$ collide) and $\Delta D[q] = w'_A$ if $q \notin \text{Fix}(S_{\text{shell}})$. Because each U_q is an $\mathbb{F}_2$-linear subspace containing 0, the inductive hypothesis $\big((p_A(t), w_A(t)) \in \Omega$ and $\forall q, D_t[q] \in U_q\big)$ holds for all $t \in [0, \infty)$ if and only if for every $(q, w_A) \in \Omega$ and every $v \in U_q$, we have $(q \oplus v \oplus w_A, P_5(v \oplus w_A)) \in \Omega$ and $\Delta D[q] \in U_q$. Exhaustive verification across all $\sum_{(q,w_A)\in\Omega} |U_q| = 2{,}528$ triples (q, w_A, v) confirms that 100% of transitions map inside Ω and U_q. Since $(0, e_R) \in \Omega$ and $D_0[q] = 0 \in U_q$ at $t = 0$, mathematical induction proves that the trajectory never leaves Ω or $\prod_q U_q$ for any $t \geq 0$, and therefore never reaches $\{5,10,21,26\}$. ■

7. Master Verification Ledger

Table 3 consolidates the mathematical theorems, computational audits, resolved questions, and open frontiers established in this work.

Status	Mathematical Claim / Compiler Result	Scope and Verification Basis
STANDS	**Even-Width Seam & Center Carrier Address Theorem** (Theorem 1.1): $R_n[h-1] = u \oplus e_R$, $R_n[h] = u \oplus e_L$, and exact closed-form addresses for $L = \text{span}\{u, e_R, e_L\}$.	Analytic proof for all $n = 2r \geq 4$; verified in Python on R_2 through R_{14}.
STANDS	**Full Symmetric Group Generation**: $\langle T, P_n, C_n \rangle = S_{2^n}$ for all $n \geq 3$, while $\langle T, P_2, C_2 \rangle \cong D_4$ (order 8) at $n = 2$.	Exhaustive group enumeration at $n = 2,3$ ($40{,}320 = 8!$) and, for $n \geq 3$, primitivity with the regular cycle $\langle T \rangle$, the fixed points of $P_n C_n$ (excluding $\text{PGL}_2(q)$) and the odd successor cycle (excluding A_{2^n});

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Status	Mathematical Claim / Compiler Result	Scope and Verification Basis
		computational checks through $n = 12$.
STANDS	**Route-Switch Defect Zeros & Boolean Möbius Shift Identity** (Theorems 2.1 & 2.2): $t_a = 0$ at $a \in \{1, 2^n - 4\}$ $(n \geq 5)$ and $MD^{(r)} = Sa^{(r)}$.	Analytic proof of the two zeros (facing law) and of the Möbius identity (Lucas' theorem); that no other zeros occur, and the $2^n - 4$ line count, computed for $n = 5..10$; Möbius identity verified with o mismatches across 77 bit planes ($n = 2..12$).
STANDS	**Carrier Subspace Confinement & Trichotomy** (Theorem 3.1): Empty-field single-reader trajectories stay inside $p_0 \oplus \text{span}\{u, w_0, J_n(w_0)\}$ with periods 6 (R_1), 132 (R_2), and 43,155,322 (R_3).	Subspace P_n-invariance proof; verified exhaustively on $n = 2..5$ and in coordinates for all widths.
STANDS	**Four-Involution Factorization & Multi-Reader Inverse** (Theorem 4.1): $F^{(m)} = \mathcal{M} \circ \mathcal{W} \circ \Pi \circ \mathcal{R}$ and $(F^{(m)})^{-1} = \mathcal{R} \circ \Pi \circ \mathcal{W} \circ \mathcal{M}$, plus conservation of shell lamps $\mathcal{L}_k$.	Exact algebraic proof for arbitrary $m \geq 1$ and simultaneous collisions; verified on 12,000 randomized states.
STANDS	**Three-Bit Affine Charge & Vacuum Separation Theorem** (Theorem 4.2): On the 8-word carrier, $Q = B_D \oplus A(p_A \oplus p_B)$ is conserved, forcing $p_A(t) \oplus p_B(t) = p_A(0) \oplus p_B(0)$ whenever $D_t = 0$.	Exact proof from $P_8(x) = A(x) \oplus u$; verified across all 64 seeds and all intermediate vacuum visits.
STANDS	**Vacuum Monodromy Closure of All 64 Same-Carrier Seeds** (Theorem 5.1): All 64 seeds close; the 28 seeds open past 300,000,000 ticks form two two-lap macro-orbits of periods 310,826,368 (16 vac) and 515,455,952 (28 vac).	Exact cocycle composition in $\mathcal{G}_{64}$ over the first-vacuum return table; 100% agreement with all 36 directly closed seeds; independently reconstructed from the census records (26 distinct periods); direct runs to completion confirm 310,826,368 for (e_R, e_L) and $(u \oplus$

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Status	Mathematical Claim / Compiler Result	Scope and Verification Basis
		$e_R, u \oplus e_L)$ and 515,455,952 for $(e_R, 0)$.
STANDS	**Cross-Shell Quadratic Partner Law, Q_{shell} Conservation, & 28-Word Confinement** (Theorems 6.1–6.3): $P_5(x) = J_5(x) \oplus (1 \oplus \ell_0)(1 \oplus \ell_1)u$; Shared Start conserves $B_D \oplus J_5(p_A \oplus p_B) = 0$ and never visits $\{5,10,21,26\}$ for $t \in [0, \infty)$.	Exact polynomial derivation and unconditional inductive closure verified across all 2,528 transitions of $(\Omega, \{U_q\})$; reproduced independently (288 reader-carry states, the stated cell dimensions, 44 field bits over 20 orbits, a 52-dimensional span, 2,528 triples, 0 failures).
RUNS	**Multi-Reader v3 Lazy-Frame Engine**: $O(1)$ transport H_h and relative-field execution $E[r] = V[r \oplus s]$ centered on reader 0.	Verified across 30,844 single-reader checks and 6,000 randomized 1-to-4-reader lazy-frame vs. absolute-field tests.
DEAD END	**1-Bit Shell Lamp $\mathcal{L}_0$ as Selector of Lap Count**: $\mathcal{L}_0$ defines a conserved sector but cannot distinguish one-lap $(B = (0, e_L), T = 26)$ from two-lap $(B = (s, e_R), T = 288)$ closures inside $\mathcal{L}_0 = 0$.	Superseded by the 3-bit affine charge Q and the $\mathcal{G}_{64}$ Vacuum Monodromy Compiler.
OPEN	**Non-Iterative Algebraic Formula for One-Lap vs. Two-Lap Selection inside $\ell_0(p_B) = 0, w_B \in R_3$**: While $\ell_0(p_B) = 1$ (32 seeds) and $w_B \in R_1 \cup R_2$ (16 seeds) are now proven 100% two-lap, a closed-form algebraic invariant separating the 8 one-lap seeds from the 7 two-lap seeds inside the 15 non-silent seeds of $\ell_0(p_B) = 0, w_B \in R_3$ without running their trajectories (up to 9,750 ticks) remains open.	Scoped to the 4×4 block $p_B \in \{0, s, u, u \oplus s\}, w_B \in R_3$.
OPEN	**Cross-Shell Vacuum Return Horizon and Asymptotic Carrier Release**: Determining the finite period $T > 2{,}000{,}000{,}000$ at which Cross-Shell Shared Start (confined to 52 affine bits) and Offset Start (122 affine bits) return to	Scoped to the 32-word two-shell subspace $W = L \oplus K$.

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Status	Mathematical Claim / Compiler Result	Scope and Verification Basis
	$D_t = 0$ and release two independent closures at shifted placements.	

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